

The arc is a good instrument; the map it draws is the room. Below 3 kHz the hub loudspeaker is a valid axisymmetric source, and from 3–5 kHz it is not.

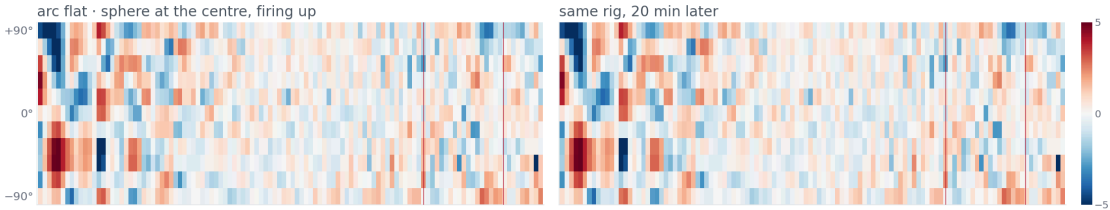
Change one thing at a time

Eleven UMIK-2 capsules on a 1.68 m arc · 112 tones, 257–6350 Hz · each cell is one capsule at one tone, dB re the eleven-capsule mean
per-capsule calibration applied · red rules mark 3–5 kHz, where the source is not axisymmetric
the sphere sits on a tripod at the centre of the arc and fires along its axis, so every capsule is the same distance away and at the same angle to it

1 Nothing changed

the same capture twice, 20 min apart

0.03 dB
r = +1.00

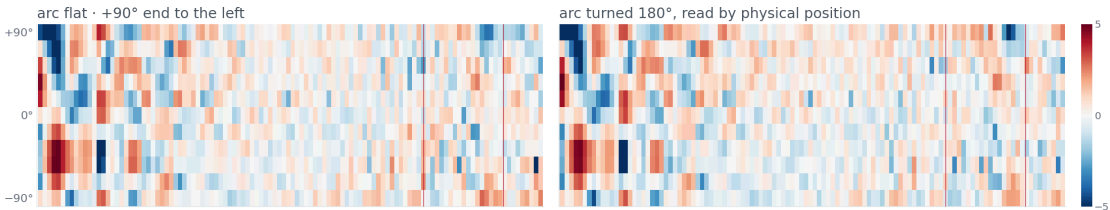


the instrument is silent when nothing moves

2 The arc flipped 180°

second run re-indexed by physical position

0.53 dB
r = +0.92

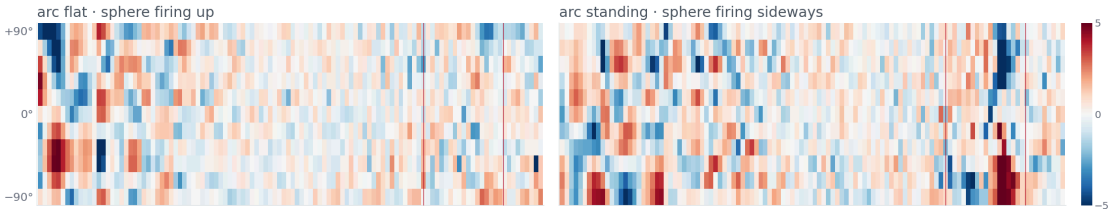


the pattern stayed with the room, not with the arc

3 The arc stood vertical

same capsules and cables, different room

2.29 dB
r = -0.12

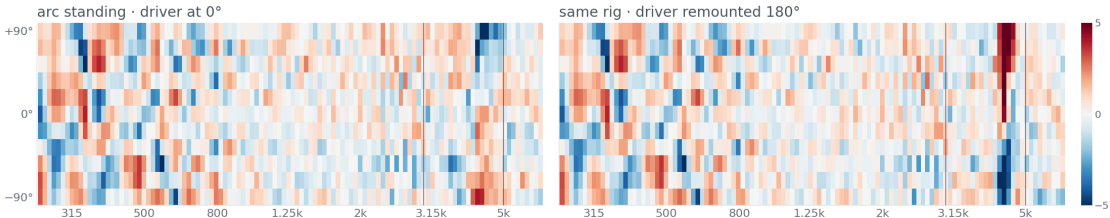


change the room and the map is unrecognisable

4 Only the driver turned 180°

in a sphere that never moved

0.20 dB
below 3 kHz



and 11.0 dB at 4362 Hz — the source's own rocking mode

0.21 dB

of the position map is provably the rig — antisymmetric under the flip

1.04 dB

is provably the room; the 0.84 dB a single flip cannot divide is room too (cov = -0.09 dB², CI [-0.22, +0.03])

≤ 0.16 dB

is all the two geometries share — so the rig's total contribution is ~0.2 dB

11.0 dB

at 4362 Hz when only the driver turns: an m = 1 rocking mode, ka = 2.39. Avoid 3–5 kHz

HOW THE CAPSULES WERE CALIBRATED

Substitution, 2026-09-16. Each of the eleven was measured in turn **at the same seat**, one capsule and one speaker in a cleared chamber, so differencing any two cancels source, room and analog chain exactly — the source only has to hold still. 97 tones, **62 Hz–16 kHz at 12/octave**, amplitude 0.03, 48 kHz, 2 s per tone and 6 s below 250 Hz; every capsule's Mic Capture Volume gated at unity first. All 55 pairs follow from the eleven captures. **Drift bracket:** 810-8904 was re-measured at the end — **-0.025 dB mean, 0.165 sd over 2 h 13 min** (0.09 dB sd below 1 kHz, 0.24 above 4 kHz), which is what licenses the method. **Seven of eleven factory files were wrong by 0.6–3.6 dB.** The datum is the four whose certificates agree with each other and with measurement (810-8900, 810-8903, 811-1896, 811-2321), not the group mean, because the error is batch-correlated. Corrections went into cal_gain_db only, held flat outside 90 Hz–16 kHz; **Sens Factors kept verbatim.** Acceptance: the spread across the eleven fell **4.01 → 0.02 dB**. Normal incidence only, which matches the arc. **No absolute anchor** — the method is blind to error common to all eleven, and only a 94 dB calibrator closes that.

OPEN, AND WHERE THE DATA IS

The loudspeaker map does not reproduce the propeller map (r = -0.04 tone-notched, +0.04 mixed; all 22 relabellings tested, best +0.24) — the propeller's own inflow, or the object at the hub, still matters. Room geometry is unrecorded: hub height, room L×W×H, foam, what stands near the arc ends. Nothing above 6350 Hz has been measured with the arc. Sessions calibrator/sessions/2026-09-16 (capsules) and 2026-09-17 (arc); levels and meta mirrored to the measurement repo, WAVs local. calibrator/rig.py captures and read_map() is the single loader, so a figure cannot disagree with the capture about what a level means; the bad band is SOURCE_SUSPECT_HZ. Every number here was recomputed from those captures while this page was written. Where this page and CLAUDE.md disagree, CLAUDE.md governs. rev 3, 2026-09-17.